a

Analysis settings

Differential expression method

DESeq2

Meta-analysis method

- Fisher
- Inverse normal

Batch effects (needs a 'batch' column in the design file)

- Batch in the DE model (~ batch + condition)
- ComBat-seq on the raw counts
- Ignore batch

Which method should I use?

Use batch-corrected values in the PCA and heatmap

Batch is handled within each study. The Batch effects tab of the results shows PCA and RLE plots before and after the correction.

RNA-Seq data

Meta-analysis of raw RNA-Seq read counts. Load the demo data, or upload a counts file and an experimental design file.

Load demo data

Raw counts file (tab-delimited)

Browse... No file selected

Experimental design file (tab-delimited)

Browse... No file selected

Download example counts file

Download example design file

The counts file has a 'GeneId' column with the gene IDs and one column of raw (unnormalised) read counts per sample, e.g. from featureCounts or htseq-count; all studies are in the same file. The design file has the columns 'sample' (the column names of the counts file), 'study' and 'condition' (and optionally 'batch'). Every condition needs at least 2 samples in a study, and every comparison at least 2 studies.

Start Analysis!

b

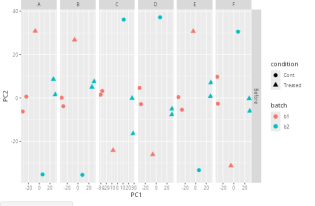
Check for batch effects (before the analysis)

Looks for batch effects in each study before the differential expression analysis: how much of the variation between samples follows the condition and how much follows the batch (if the design file has a 'batch' column), with PCA and RLE plots. Use it to decide on the batch setting above.

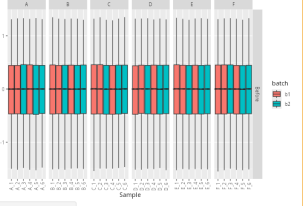
Check for batch effects

Study	Samples	Variance explained by condition (%)	Variance explained by batch (%)	Advice
A	6	28	31	Batch explains as much variance as condition or more (31% vs 28%): consider 'Batch in the DE model' or ComBat-seq.
B	6	27	31	Batch explains as much variance as condition or more (31% vs 27%): consider 'Batch in the DE model' or ComBat-seq.
C	6	26	32	Batch explains as much variance as condition or more (32% vs 26%): consider 'Batch in the DE model' or ComBat-seq.
D	6	28	32	Batch explains as much variance as condition or more (32% vs 28%): consider 'Batch in the DE model' or ComBat-seq.
E	6	29	32	Batch explains as much variance as condition or more (32% vs 29%): consider 'Batch in the DE model' or ComBat-seq.
F	6	28	32	Batch explains as much variance as condition or more (32% vs 28%): consider 'Batch in the DE model' or ComBat-seq.

PCA of each study before the analysis



Relative log expression before the analysis



Download PCA

Download RLE plot

c

Microarray data

Meta-analysis of normalised microarray expression data. Load the demo data, upload one expression file per study with a design file, or import series from GEO.

Load demo data

Data source

- Upload expression files
- Import from GEO

GEO series accessions, separated by commas

GSE63352

Fetch from GEO

For each series, choose the column that holds the condition and give each of its values a condition label. Samples whose value has no label are not used. Use the same labels in all series (e.g. 'Control', 'Treated').

GSE63352

Condition column

characteristics_ch1

Gene column

None (use the probe IDs)

genotype/variation: WT (6 samples)

condition label, or empty to leave out

genotype/variation: ΔndhR (Δslr1594::Km) (7 samples)

condition label, or empty to leave out

Start Analysis!

d

Scope

TMA does a **gene-level** meta-analysis of **bulk RNA-seq** (raw read counts per gene) and of **single-colour microarray** data (normalised expression per probe or gene). In each study, every condition is compared with a reference condition; the p-values of the studies are then combined for each gene.

- Transcript-level, isoform and alternative-splicing analyses are out of scope: the input must be one value per gene (RNA-seq) or per probe that is collapsed to its gene (microarrays).
- RNA-seq and microarray studies are analysed in separate tabs and cannot be combined in one meta-analysis.
- Two-colour microarrays are not supported, because their channels are not comparable as independent samples.

Study requirements

- **At least 2 samples (replicates) per condition in a study.** A condition with fewer samples is left out of that study, and the study is not used for the comparisons that involve it.
- **At least 2 studies per comparison.** A comparison with fewer studies is not meta-analysed. If no comparison can be meta-analysed, the analysis stops and says why.
- **Gene-level IDs.** The same type of gene ID in every study, so that genes can be matched across studies. For microarrays, the probes are collapsed to genes (the probe with the highest mean expression is kept) and only the genes measured in every study are used.
- **Supported ID types for the enrichment:** see the table below. The app reports, for every enrichment, how many IDs the service recognised.

ID type	PANTHER (GO)	KEGG	STRING
Gene symbol	yes	no	yes
Locus tag / KEGG gene ID	yes	yes	yes
NCBI Entrez Gene ID	yes	yes	yes
UniProt accession	yes	yes	yes
Ensembl gene ID	yes	no	yes